

Appendix Table 4 - Houston Achievement Specification Tests - Grade Level Evacuee Share, Elementary

		A. TAKS Math					B. TAKS Reading				
Pre- Katrina TAKS Quartile →		All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile	All	Bottom Quartile	2nd Quartile	3rd Quartile	4th Quartile
Model ↓											
(1)	IV (Shelter Instrument Only)	0.76 (1.34)	-0.51 (2.24)	-0.36 (1.38)	-1.05 (1.07)	0.33 (0.79)	1.28 (1.35)	0.48 (2.44)	1.75 (1.73)	0.09 (0.76)	0.45 (0.89)
(2)	IV (Shelter & 9/13/05 Instruments)	-0.47 (0.76)	-1.09 (1.31)	-1.71 (1.05)	-1.32 (0.96)	0.54 (0.73)	-0.27 (0.83)	-0.57 (1.81)	1.15 (1.07)	0.34 (0.66)	1.19 (0.75)
(3)	Student & School Fixed Effects	-0.49 (0.34)	0.02 (0.66)	-0.73 (0.45)	-0.77 (0.62)	0.05 (0.57)	0.32 (0.36)	0.39 (0.76)	-0.08 (0.57)	0.82 (0.53)	0.69 (0.57)
	Observations	90,267	20,229	23,183	23,368	23,487	73,923	16,813	19,348	18,977	18,785
(4)	Class Level Katrina Share (2003-04 through 2005-06 only)	-0.19 (0.26)	0.02 (0.39)	-0.86*** (0.26)	-0.84** (0.37)	-0.14 (0.30)	0.50* (0.26)	0.04 (0.51)	0.61* (0.33)	0.39 (0.34)	0.12 (0.38)
	Observations	79,626	17,628	20,524	20,700	20,774	65,791	14,860	17,309	16,915	16,707
(5)	Limited to Schools with < 10% Katrina in 0506	-0.19 (0.71)	-1.32 (1.23)	-1.36* (0.81)	-0.72 (0.77)	0.84 (0.84)	-0.38 (0.65)	-0.76 (1.32)	-0.17 (0.88)	-0.02 (0.60)	-0.37 (0.94)
	Observations	88,722	19,832	22,761	22,981	23,148	72,617	16,430	18,986	18,682	18,519
(6)	Interrupted Panel - 2003-04, 05-06, and 06-07 Only	-0.20 (0.53)	-0.41 (0.97)	-1.04 (0.66)	-0.83 (0.62)	1.05** (0.53)	-0.16 (0.61)	-0.16 (1.12)	0.13 (0.95)	-0.55 (0.55)	0.47 (0.66)
	Observations	63,797	14,609	16,307	16,397	16,484	46,779	10,969	12,235	11,878	11,697
	Instrument for Evac Share w/ Evac Share in 0405 School										
(7)	OLS	-0.37 (0.45)	-0.88 (0.91)	-1.15** (0.54)	-1.25** (0.62)	0.69 (0.53)	-0.33 (0.43)	-0.31 (0.91)	-0.05 (0.65)	-0.08 (0.42)	0.04 (0.58)
	Second Stage	-0.23 (0.48)	-0.71 (1.15)	-1.14** (0.57)	-1.12* (0.63)	0.60 (0.53)	-0.41 (0.48)	-0.78 (1.36)	-0.06 (0.71)	-0.19 (0.50)	0.20 (0.58)
	Observations	78,995	17,445	20,357	20,531	20,662	65,179	14,679	17,144	16,771	16,585
(8)	Unstandardized Scale Scores	-134 (123)	-187 (215)	-297* (152)	-321* (175)	109 (123)	-124 (122)	-201 (260)	-77 (138)	-8 (98)	93 (145)
	Observations	90,292	20,297	23,166	23,269	23,560	73,938	16,753	19,389	18,994	18,802
(9)	Replace Students Enrolled But Missing Score with Lagged Score	-0.22 (0.39)	-0.60 (0.76)	-1.14** (0.49)	-0.91 (0.56)	0.80 (0.49)	-0.13 (0.39)	-0.56 (0.91)	0.06 (0.55)	0.11 (0.37)	0.23 (0.55)
	Observations	92,167	21,306	23,417	23,636	23,808	75,893	17,898	19,540	19,299	19,156
(10)	Replace Students Enrolled But Missing Score with Mean Score for Bottom Quartile Students	-0.41 (0.41)	-0.98 (0.78)	-1.16** (0.50)	-0.96 (0.63)	0.24 (0.47)	-0.04 (0.39)	-0.21 (1.00)	0.05 (0.56)	-0.18 (0.38)	0.26 (0.56)
	Observations	92,250	21,355	23,434	23,646	23,815	76,036	18,006	19,560	19,307	19,163
	Standardize Using District Means and Standard Deviations for 2004-05 Cohort										
(11)	Grade Level Katrina Share	-0.15 (0.23)	-0.20 (0.47)	-0.67** (0.29)	-0.49 (0.30)	0.32 (0.28)	-0.01 (0.21)	0.15 (0.43)	-0.00 (0.26)	0.05 (0.18)	0.04 (0.28)
	Observations	90,267	20,301	23,143	23,277	23,546	73,923	16,740	19,388	18,990	18,805
(12)	School - Year Fixed Effects (2006, 2007 Only)	-0.38 (0.82)	0.06 (1.34)	-0.82 (1.03)	-0.54 (0.98)	1.77 (1.52)	0.88 (1.04)	2.67 (1.88)	1.19 (1.17)	0.87 (1.27)	-1.17 (1.25)
	Observations	36,113	8,788	9,048	9,082	9,195	30,866	7,294	7,783	7,851	7,938
(13)	School - Grade Fixed Effects	-0.15 (0.47)	-0.56 (0.89)	-1.06* (0.55)	-0.94 (0.62)	0.46 (0.47)	-0.11 (0.43)	-0.37 (1.02)	0.10 (0.67)	-0.01 (0.40)	0.21 (0.55)
	Observations	90,267	20,229	23,183	23,368	23,487	73,923	16,813	19,348	18,977	18,785
(14)	Evacuee Counts in Grade with Quartic Control in Grade-Level Enrollment	-0.0020 (0.0038)	-0.0069 (0.0080)	-0.0069* (0.0041)	-0.0065 (0.0047)	0.0058 (0.0043)	-0.0004 (0.0035)	-0.0007 (0.0075)	0.0024 (0.0047)	0.0034 (0.0037)	0.0038 (0.0039)
	Observations	90,267	20,229	23,183	23,368	23,487	73,923	16,813	19,348	18,977	18,785
	Separate Estimates for New Entrants to School										
(15)	Evacuee Share in Grade	-0.21 (0.42)	-1.22 (0.97)	-0.99** (0.50)	-0.73 (0.55)	0.93* (0.50)	-0.03 (0.41)	-0.20 (1.10)	0.27 (0.59)	0.21 (0.41)	0.39 (0.55)
	Evacuee Share in Grade X New Entrant	-0.37 (0.55)	1.93 (1.23)	-0.94 (0.70)	-1.89* (1.00)	-3.40** (1.58)	-0.59 (0.52)	-1.00 (1.34)	-0.88 (0.91)	-0.42 (0.90)	-2.46 (1.58)
	Observations	90,267	20,229	23,183	23,368	23,487	73,923	16,813	19,348	18,977	18,785
(16)	Exclude Schools With No Evacuees in 2005-06	-0.19 (0.41)	-0.57 (0.87)	-0.92* (0.49)	-0.88 (0.58)	0.67 (0.50)	0.07 (0.41)	-0.40 (0.96)	0.39 (0.58)	0.18 (0.40)	0.22 (0.57)
	Observations	79,203	17,575	20,333	20,465	20,830	65,181	14,581	16,756	16,725	17,119

Standard errors are provided in parentheses and clustered by school. Regressions cover 2003-04 - 2006-07 except where noted and include student's race, gender, free/reduced price lunch status, grade-by-year fixed effects, school fixed-effects, and lagged achievement from the student's most recent pre-Katrina lagged score interacted with indicators for the number of years between exams. TAKS scores are standard deviations of scale scores within grade and year using statewide means and standard deviations. When students have multiple scores for a single subject in a given year we use the lowest score. Elementary covers students in grades 3 - 5 prior to 2006, grades 4 & 5 in 2006, and grade 5 in 2007. Middle/High is any student in grades 6 - 11. *, **, and *** reflect significance at the 10%, 5%, and 1% levels, respectively.